Kenneth Wraight

Counter-Narratives PG Conference 12/06/2025

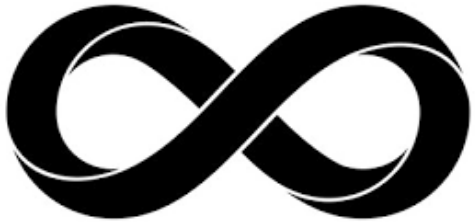
Synergistic Studies:

Connecting Historical Games Studies and Science and Technology Studies





- Motivation
- STS & Synergies
- HGS & Narrative Hegemony
- Crossings Between
- Summation



Motivation

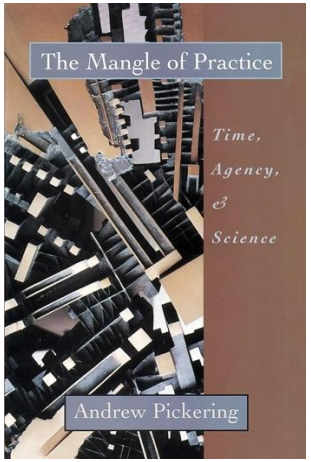
The Historical Researcher

- Fulltime researcher in Experimental Particle Physics in Physics & Astronomy
- Parttime MPhil(res) post-graduate student with History



The Historical research:

- A nineteenth century debate: the Age of the Earth
  - (see back-up)
- Applying an analytical framework: The Mangle by Andrew Pickering
- Goals: textual dissertation and a pedogeological game



Along the way, I stumbled upon interesting parallels between two separated disciplines:

- **Science and Technology studies** (old friend)
- **Historical Game Studies** (new acquaintance)

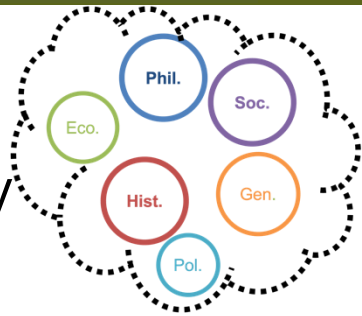
→ **To what extent are these studies synergetic?**

⚠ "Science" will be used with *constructive ambiguity* and include Technology

- Roughly: official knowledge of/tools for the Natural World



STS Overview



Science and Technologies (STS) is a group of sub-disciplines with common interest in investigating Science

- *Multi-/trans-/meta-discipline*: inc. philosophy, history, sociology

Activities officially grouped together in the 1970s

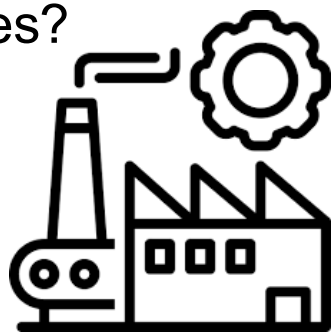
- **NB** much work had taken place previously as independent pursuits (e.g. Frankfurt school, Michael Polanyi, Thomas Kuhn, etc.)

The activities overlap in areas of interest, methodologies and analyses

- including investigating the ***hegemonic narrative* of Science**

Roughly: How is Science made and what are the consequences?

- Who makes Science and how?
- Could Science have been different?
- How does Science interact with Society?
- Who is affected by Science and how?



Investigations *foreground participants and practice* over (traditionally favoured) theories and concepts

Synergies

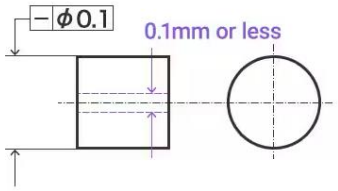
STS shares several themes with other interdisciplinary activities

- *Multiple perspectives with flexible relationships*

Some themes with STS examples…

Tolerance in the object: non-hegemonic voices in Science

- Official science (e.g. physics), para-science (e.g. psychics), archaic science (e.g. astrology)

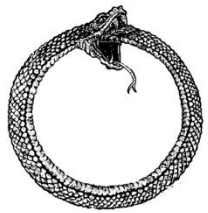


Tolerance in the subject: internal disputes in STS

- Grouping of disciplines with potentially conflicting views:
  e.g. philosophy (normative), sociology (descriptive), social theory (critical)

Antagonisms with parent discipline or discipline(s) of interest

- Indifference: STS is not a standard part of UG courses in sciences
- Animosity: cf. Sokal affair



Reflexivity:

- STS as an object of STS: concerns role of “Society” in STS descriptions



HGS Overview

Historical Game Studies (HGS) is a grouping of interests around engagement with History through games

- *Multi-/trans-/meta-discipline*: inc. history, media, memory, technology

Group coalesced in the 2000s

- **NB** Work had taken place previously as independent research areas

The activities overlap in areas of interest, methodologies and analyses

- including investigating the ***hegemonic narrative* of History**

Roughly: how is historical engagement made through games and what are the consequences?

- Who makes historical narratives and how?
- How is historical content communicated through games?
- How does this gamified historical engagement interact with Society?
- Who is engaged and how?

Investigations *foreground participants and engagement* over historical accuracy

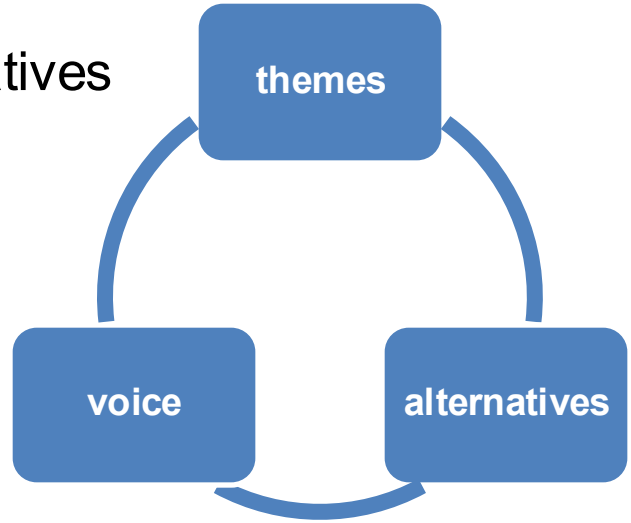
Highlighting Hegemony

How is a narrative created?

Games can *invent, support or contest* historical narratives

- Topics/themes present the user with a “real” story
- Characters and scenes provide voice/visibility to communities and events
- Alternatives highlight causal networks

TL;DR How is history used?



What are the consequences?

- Interpretations of narrative
- Public/shared perception of history
- Public/shared ways to remember

TL;DR What is engagement?

"games that in some way represent the past or relate to discourses about it, the potential applications of such games to different domains of activity and knowledge, and the practices, motivations and interpretations of players of these games and other stakeholders involved in their **production or consumption"**



Crossing: HGS → STS

Many of the themes of HGS concerning hegemonic narrative creation and engagement are echoed in STS.

Example hegemonic narrative of Science (strawman?):
Science is the method of establishing facts about the Natural World

- Homogenous, unique, rational, deterministic

How is a narrative created?
Science can *invent, support or contest* participant interests

- Participants produce (e.g. contest/dispute/establish) a “true” story
- Participants do not necessarily reflect voices in Society
- Examples of disputes/failure, as well as success, show method

TL;DR How do people produce (scientific) knowledge?

What are the consequences?

- Interpretations of narrative
- Public/shared perception of Nature
- Public/shared ways to do science and be scientific

TL;DR What is role of science in society?

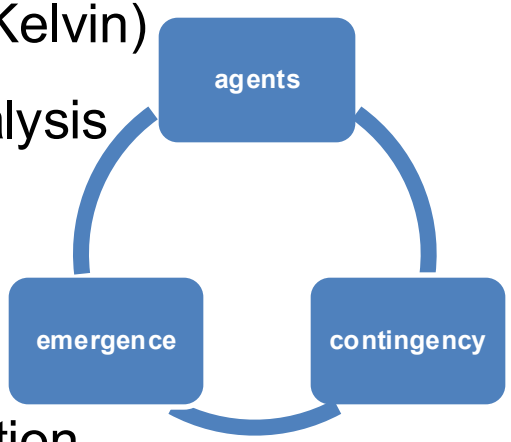


Crossing: STS → HGS

One of the outcomes of my research is to create a pedogeological game

Aiming to communicate some of the themes of research

- STS analysis: *contingency, agency and emergence* in Scientific practice
- History: *local* significance of geochronological debate (Kelvin)



Game development/analysis itself is amenable to STS analysis

- **Agency: *responses*** *to events*
  - Human agents: player actions
  - Non-human agents: simulated players, game rules
- **Contingency: *novelty* in repeated plays**
  - Alternatives in game outcome: victory, path or duration
- **Emergence: some (game-changing) element of *chance***
  - Events not controlled by agents, which can affect the game outcome

NB Engaging participants in a game with STS themes deviates from traditional hegemonic narratives presented in Historical Science games



Analytical Examples

Illustrate analytical themes with a few examples of games

- **Snakes & Ladders** (🐍🐍&☐ ☐)
  - *Agency*: no human agency (H) in dice rolls; non-human agency (N) in game rules when player lands on snake or ladder
  - *Contingency*: chance in di(c)e rolls makes games novel
  - *Emergence*: uncontrolled positioning of snakes and ladders
- **Dominos** (🁣🁥🁫):
  - *Agency*: players choose what to play (i.e. skill); no game agency
  - *Contingency*: chance in initial tile allocation
  - *Emergence*: no non-agent events
- **Monopoly** (🎩🐕🧐):
  - *Agency*: skill in purchases; game agency in rules
  - *Contingency*: chance in dice rolls and player choice
  - *Emergence*: chance cards add uncontrolled events

	🐍🐍&☐ ☐	🁣🁥🁫	🎩🐕🧐
Agency (H,N)	✗, ✓	✓, ✗	✓, ✓
Contingency	✓	✓	✓

Summary

Goal has been to relate meta-studies from different academic communities:

- Science and Technology Studies (STS)
- Historical Game Studies (HGS)

To what extent are these studies synergetic?

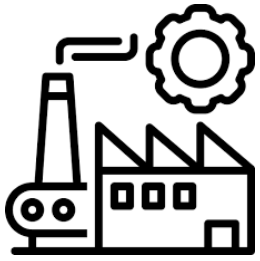
- **Very!**

Already share several themes:

- Narrative building
- Consequences of engagements
- Tolerant to varying interests with shared methods

Fruitful opportunities are available for further engagement

- Developing existing synergies
- Bringing STS analysis to HGS







University of Glasgow

Back-up

A Brief (Geo)History of Time I

Geology by the nineteenth century

Separation of the history of the Earth from the (biblical) history of Man.

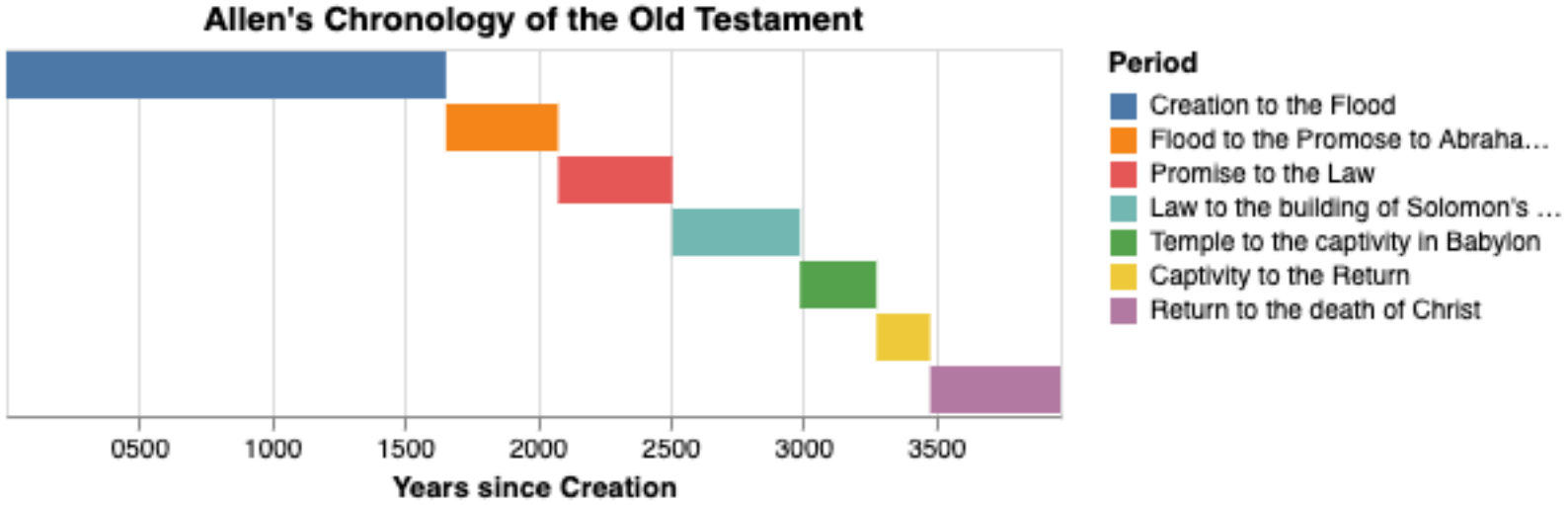
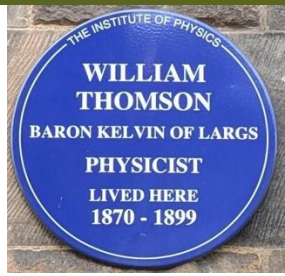


Figure 1: Allen's estimates of the Age of the Earth based on Biblical exegesis. The timeline runs from Creation (LHS) to Christ's Death (RHS), after which calendar years may be added to the present. Data from Jackson (2018), chpt. 2

Geology becoming:

- **Professional**: academic positions, journals, texts
- **Secular**: leave biblical interpretation behind
- **Empirical**: practitioners around the (colonised) world.

A Brief (Geo)History of Time II



Nineteenth Century Debate

Hutton, Lyell (et al) considered the age of the Earth to be infinitely / indeterminately long in processes of denudation and deposition: "we find no vestige of a beginning, no prospect of an end"

This offended Kelvin – thermodynamics does not do infinite

Over mid to late nineteenth century a debate ensued.

By the end of the century most estimates fell within Kelvin's **100 Myear** limit

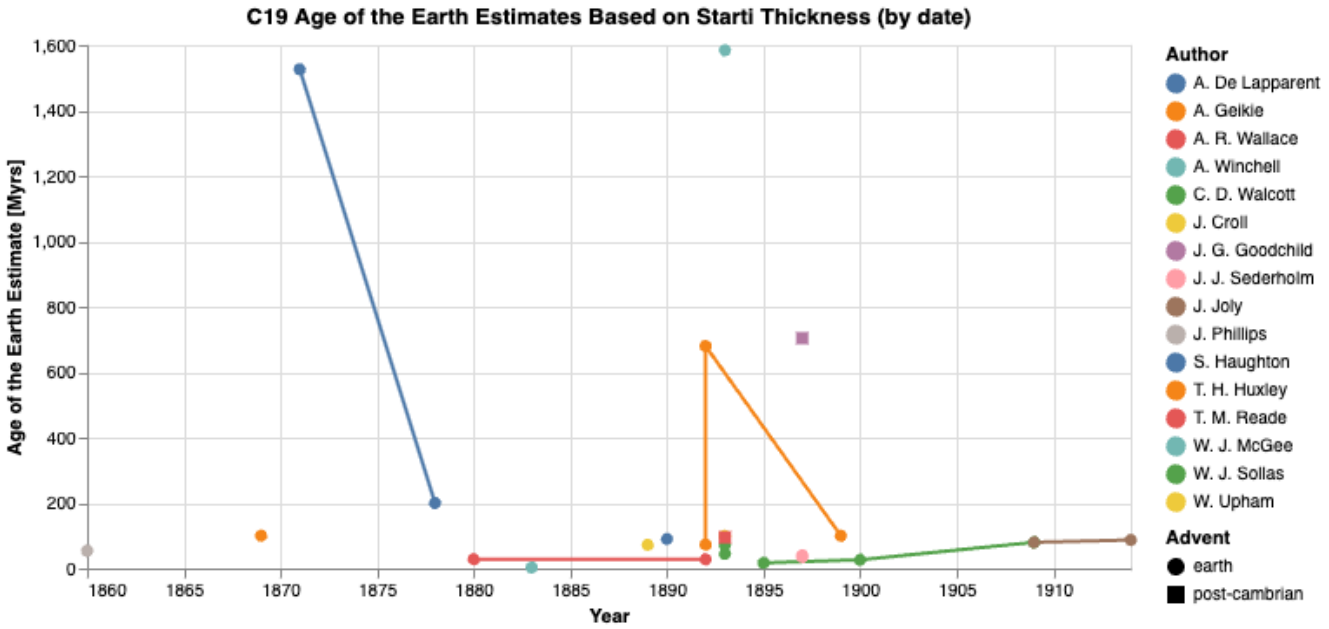


Figure 2: A selection of Nineteenth century estimates of the Age of the Earth after Thomson's 1862 paper (by author). Most estimates fit roughly into the 100 million year limit. Data from Jackson (2018), chpt. 10.

A Brief (Geo)History of Time III

Beyond the Debate

Early in twentieth century radioactivity was discovered

After a couple of decades decay chains were established as best source of geochronological measure

Arthur Holmes work established new limit: **~3.3Byears**

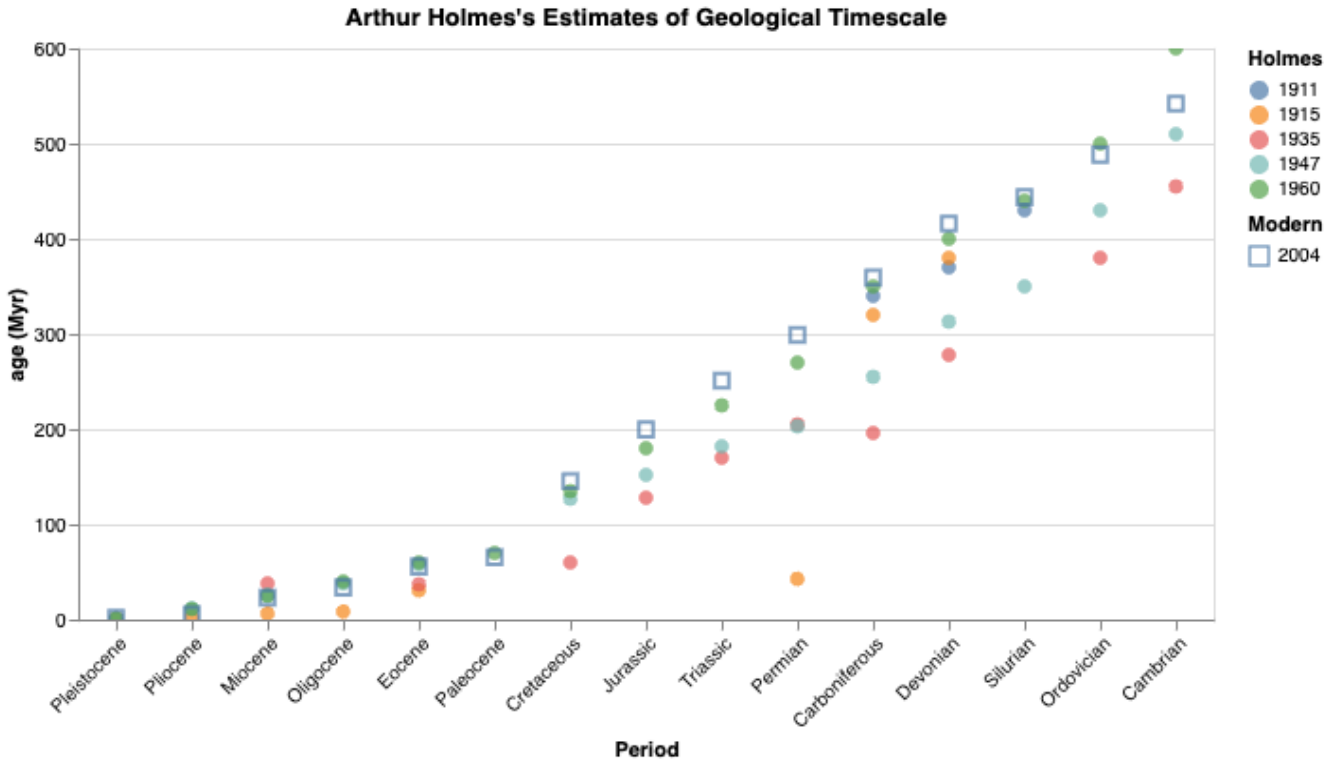


Figure 3: Arthur Holmes's estimates of the duration of Stratigraphical Periods based on radioactive dating. By 1960 his results had almost achieved the modern standard. Data from Jackson (2018), chpt. 13.

A Brief (Geo)History of Time IV

Modern Perspective

Kelvin model (1860s): cooling Earth, cooling Sun, rigid Earth

- Terrestrial heat from radioactivity: ~1910s
- Sun as nuclear fusion engine: ~1950s
- Molten mantle beneath Earth's solid crust: ~1960s

Radiometric dating of oldest meteors → **4.5Byears** age of the Earth

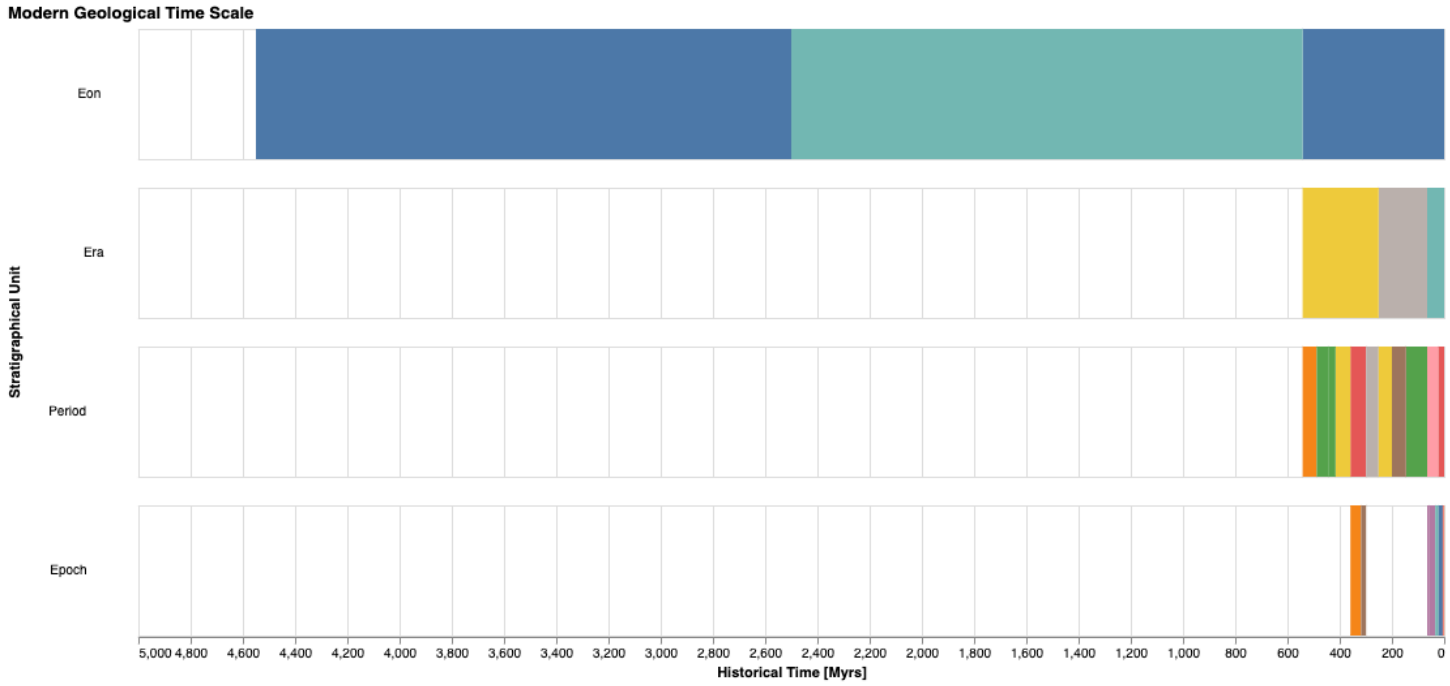


Figure 5: Modern Chronology of the Earth. The Geological Column (here presented horizontally) with a chronological timescale. Data from Jackson (2018), cover page.